

Developer Mock Test Results

Test ID:	d15cef3e-1e66-41...	Student ID:	9858
Test Type:	Developer Assessment	Date:	2026-02-17 15:19:54
Overall Score:	0/25 (0.0%)	Performance:	■■ Needs Improvement
Warnings:	0/3	Status:	Completed ■

Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	0/10	0.0%	■■ Needs Work
■ MCQ/Theory	0/10	0.0%	■■ Needs Work
■ Coding	0/5	0.0%	■■ Needs Work

Detailed Question Review

APTITUDE SECTION (0/10 - 0.0%)

■ Q1. A shirt is on sale for 10% off. If the original price is Rs.500, how much will you pay?

Your Answer: Rs.500

Correct Answer: Rs.450

■ To find the sale price, you need to calculate 10% of the original price, which is Rs.500, and then subtract that amount from the original price. The correct calculation is 10% of Rs.500 = $0.1 \times 500 = \text{Rs.}50$, so the sale price is $\text{Rs.}500 - \text{Rs.}50 = \text{Rs.}450$. It seems like there was a mistake in calculating the discount, but don't worry, it's an easy concept to learn and practice, and you'll get the hang of it with a little more practice.

■ Q2. A can do a piece of work in 8 days. B can do the same work in 10 days. C can do the same work in 12 days. How many days will it take for A, B, and C t...

Your Answer: __SKIPPED__

Correct Answer: 4 days

■ To find the time it takes for A, B, and C to do the work together, we calculate the fraction of work each can do in a day and add them together: A can do $\frac{1}{8}$ of the work, B can do $\frac{1}{10}$, and C can do $\frac{1}{12}$, so together they can do $(\frac{1}{8} + \frac{1}{10} + \frac{1}{12})$ of the work in a day, which simplifies to $\frac{1}{4}$, meaning they can complete the work in 4 days. If you skipped this question or got it wrong, you might have misunderstood how to combine their work rates or made a calculation error, but that's okay - it's all part of the learning process. Keep practicing and you'll get better at solving these types of problems, and don't hesitate to ask for help if you need it.

■ Q3. What is the next number: 1, 4, 9, 16, 25, ?

Your Answer: __SKIPPED__

Correct Answer: 36

■ The correct answer is 36 because the pattern of the sequence is obtained by squaring the consecutive integers, where $1^2 = 1$, $2^2 = 4$, $3^2 = 9$, $4^2 = 16$, $5^2 = 25$, and $6^2 = 36$. If you chose a different option, you might have misidentified the

pattern or made a calculation error. Don't worry, it's all part of the learning process, and recognizing patterns in sequences is a valuable skill that takes practice to develop.

■ **Q4. The average of 3 numbers is 15. What is the sum of the 3 numbers?**

Your Answer: __SKIPPED__

Correct Answer: 45

■ To find the sum of the 3 numbers, we multiply the average by the total number of values, so $15 * 3 = 45$. This calculation makes option C the correct answer, as it accurately represents the sum of the three numbers. If you chose a different option, you might have misunderstood the concept of average or made a calculation error, but don't worry, practicing with similar problems will help you build confidence and accuracy in your math skills.

■ **Q5. What is the next number: 1, 2, 4, 7, 11, ?**

Your Answer: __SKIPPED__

Correct Answer: 15

■ The correct answer is $16 - 1 = 15$ because this sequence is formed by adding 1, 2, 3, 4, and so on to the previous term. If you chose a different option, you might have misidentified the pattern or added the wrong number to the previous term, but that's okay - recognizing patterns in sequences takes practice. Keep working on it, and you'll get better at identifying these types of sequences and making accurate predictions.

■ **Q6. A bag contains 5 red balls and 3 blue balls. What is the ratio of red balls to blue balls?**

Your Answer: __SKIPPED__

Correct Answer: 5:3

■ The correct answer is 5:3 because it directly represents the number of red balls to the number of blue balls in the bag. To find the ratio, you simply compare the number of red balls, which is 5, to the number of blue balls, which is 3, so the ratio is 5:3. If you chose a different answer, you might have simplified the ratio incorrectly or miscounted the number of balls, but don't worry, it's an easy concept to understand with a little practice, so keep trying and you'll get the hang of it.

■ **Q7. A shopkeeper buys 100 items for Rs.100. He sells 50 items for Rs.60. What is his profit percentage?**

Your Answer: __SKIPPED__

Correct Answer: 10%

■ The correct answer is 10% because the shopkeeper's cost price for 50 items is Rs.50, and he sells them for Rs.60, making a profit of Rs.10. If you skipped this question or got it wrong, you might have miscalculated the cost price per item or forgotten to calculate the profit percentage, but don't worry, it's an easy concept to grasp with practice. To find the profit percentage, simply divide the profit by the cost price and multiply by 100, and you'll get the correct answer, which in this case is 10%.

■ **Q8. There are 5 houses in a row, each painted a different colour: blue, green, red, white, and yellow. Each house is occupied by a person of a different n...**

Your Answer: White

Correct Answer: Blue

■ The correct answer is blue because the Canadian lives in the first house, and since the American lives in the red house, the Canadian cannot live in the red house, and through the process of elimination with the other clues, we can deduce that the Canadian lives in the blue house. It's likely that you made a mistake by not carefully considering the order of the houses and the given clues, which led you to incorrectly choose white as the colour of the Canadian's house. Don't worry, it's all part of the learning process, and with practice, you'll become more skilled at solving these types of puzzles.

■ **Q9. There are 3 switches, but they are not labelled. Each switch corresponds to one of three light bulbs in a room. Each light bulb is either on or off. Y...**

Your Answer:

__SKIPPED__

Correct Answer:

Turn switch 1 to on for 5 minutes, then turn it off. Turn switch 2 to on and enter the room.

■ The correct answer works because turning switch 1 to on for 5 minutes allows the corresponding bulb to heat up, then turning it off and turning switch 2 to on means you can identify two bulbs: the hot one that was recently turned off and the one that's currently on. If you skipped the question or chose a different option, you might have missed the importance of

using the heat of the bulb as a clue, or not considered how to uniquely identify each bulb with just one observation. By thinking creatively about how to use the switches and the bulbs' physical properties, you can solve this puzzle and develop your problem-solving skills.

■ **Q10. A can do a piece of work in 10 days. B can do the same work in 15 days. How many days will it take for A and B to do the work together?**

Your Answer: __SKIPPED__

Correct Answer: 6 days

■ To find the answer, we calculate the work done by A and B together in one day, which is $1/10 + 1/15 = 3/30 + 2/30 = 5/30 = 1/6$. This means they complete $1/6$ of the work in one day, so they will finish the work in 6 days. If you chose a different answer, you might have made a calculation error when finding the combined work rate of A and B, but don't worry, it's a great opportunity to practice and improve your problem-solving skills.

■ MCQ SECTION (0/10 - 0.0%)

■ **Q1. What is the purpose of the `__init__` method in a Python class?**

Your Answer: To reduce memory usage

Correct Answer: To assign values to instance variables when an object is created

■ The correct answer is right because the `__init__` method in a Python class is a special method that is automatically called when an object of that class is created, allowing you to assign initial values to instance variables. It seems you were mistaken, as reducing memory usage is not the primary purpose of the `__init__` method, and this confusion might have arisen from not fully understanding the method's role in object initialization. Don't worry, it's a common learning point, and now you know the correct purpose of the `__init__` method, which will help you in creating and initializing objects in your Python programs.

■ **Q2. What are the fundamental concepts that Python programs are built upon?**

Your Answer:

Objects, classes, and inheritance

Correct Answer:

Variables, data types, control structures, and functions

■ Python programs are built upon fundamental concepts such as variables, data types, control structures, and functions, which provide the foundation for writing executable code and controlling the flow of a program. The user's answer, focusing on objects, classes, and inheritance, is related to object-oriented programming, a key aspect of Python, but it's not the most basic building blocks of the language. By starting with the basics, you'll have a solid understanding of how Python works, and then you can move on to more advanced concepts like object-oriented programming, so don't worry, you're on the right track, just take it one step at a time.

■ **Q3. What is the correct step to write a Python program?**

Your Answer: __SKIPPED__

Correct Answer: Install Python, open a code editor, write code, run the program

■ The correct answer is the right sequence because you need to have Python installed on your computer to run Python programs, then you can open a code editor to write your code, and finally run the program to see the results. If you skipped this question or got it wrong, you might have misunderstood the order of steps or thought that you could run a program without having Python installed first. Don't worry, it's an easy mistake to make, and now you know the correct steps to write and run a Python program, so keep practicing and you'll become more comfortable with the process.

■ **Q4. What is the purpose of the `__init__` method in Python classes?**

Your Answer: __SKIPPED__

Correct Answer: To initialize objects with instance variables

■ The correct answer is right because the `__init__` method in Python classes is a special method that is automatically called when an object of the class is created, allowing you to set initial state of the object by assigning values to instance variables. If you skipped this question, you might have misunderstood the role of `__init__`, thinking it's for defining the class or creating objects, but it's actually for setting up the object's initial state. Don't worry, it's an easy concept to grasp, and with practice, you'll become more comfortable with using `__init__` to initialize objects in your Python classes.

■ Q5. What is the first step in writing a Python program?

Your Answer: __SKIPPED__

Correct Answer: Install Python on your system

■ To start writing a Python program, you first need to have Python installed on your system, which is why installing Python is the correct first step. If you skipped this question or chose a different option, you might have been eager to dive into coding, but it's essential to set up your environment first. Don't worry, it's an easy mistake to make, and now you know the crucial first step to begin your Python programming journey.

■ Q6. What are the four fundamental principles of Object-Oriented Programming (OOP) in Python?

Your Answer: __SKIPPED__

Correct Answer: Encapsulation, abstraction, inheritance, and polymorphism

■ The correct answer is encapsulation, abstraction, inheritance, and polymorphism because these four principles form the foundation of Object-Oriented Programming (OOP) in Python, allowing for the creation of reusable, modular, and efficient code. If you skipped this question or chose a different option, you might have gotten confused with other important concepts in Python, such as control structures or file handling, which are not directly related to OOP principles. Don't worry, it's all part of the learning process, and understanding these fundamental principles will help you become proficient in Python programming.

■ Q7. What is the purpose of using try-except blocks in Python?

Your Answer:

__SKIPPED__

Correct Answer:

To handle errors and exceptions that may occur during program execution

■ The correct answer is right because try-except blocks in Python are specifically designed to catch and manage errors or exceptions that may arise during the execution of a program, allowing it to continue running smoothly. If the user skipped this question, they might have been unsure about the purpose of try-except blocks, possibly thinking they were used for optimization or code organization, which are important but not the primary function. By understanding the role of try-except blocks, you can write more robust and reliable code, so don't be discouraged if you weren't sure - it's all part of the learning process.

■ Q8. What is the primary concept of Object-Oriented Programming (OOP) in Python?

Your Answer: __SKIPPED__

Correct Answer: Objects and classes

■ The primary concept of Object-Oriented Programming (OOP) in Python is indeed objects and classes, as they are the fundamental building blocks that allow for the creation of reusable and modular code. If you skipped this question or chose a different option, you might have been focusing on other important aspects of Python, such as functions, variables, or control structures, but not realizing that OOP is specifically centered around objects and classes. Don't worry, it's all part of the learning process, and understanding OOP concepts will help you become a more proficient Python programmer.

■ Q9. What is the purpose of the 'a' mode in Python file handling?

Your Answer: __SKIPPED__

Correct Answer: To add data to the end of a file

■ The 'a' mode in Python file handling is used to add data to the end of a file, making it the correct answer. This is because when a file is opened in 'a' mode, the cursor is placed at the end of the file, allowing new data to be appended without overwriting existing content. If you were unsure, you might have confused 'a' mode with 'w' mode, which overwrites existing content, but don't worry, practice and review will help you master Python file handling modes.

■ Q10. What is the purpose of method overriding in Python classes?

Your Answer: __SKIPPED__

Correct Answer: To allow child classes to provide their own implementation of a method

■ The correct answer is right because method overriding in Python classes enables child classes to define their own version of a method that is already present in the parent class, allowing for more specific or customized behavior. This is a fundamental concept in object-oriented programming, which is why option D, although related, is not the most precise answer. If you skipped this question, don't worry - it's a great opportunity to learn and understand how method overriding helps create more flexible and reusable code in Python.

■ CODING SECTION (0/5 - 0.0%)

■ **Q1. Write a Python program to demonstrate error handling using try-except blocks. ****Input:**** A list of numbers (e.g., `[1, 2, 3, "four", 5]`) ****Output:**** ...**

Your Answer:

wergthgfds

Correct Answer:

```
def process_numbers(numbers):
    try:
        for num in numbers:
            if not isinstance(num, (int, float)):
                raise ValueError("Invalid input")
            print(f"Processing number: {num}")
    except ValueError as e:
        print(f"Error: {e}")

numbers = [1, 2, 3, "four", 5]
process_numbers(numbers)
```

■ It looks like there's a bit of a problem with your code - it seems you haven't actually written any code, and instead have a string of random characters. Don't worry, we can start fresh. The correct code works because it uses a try-except block to catch any `ValueError` exceptions that are raised when it encounters a non-numeric value in the list, allowing it to print a helpful error message instead of crashing. By using this approach, you can make your code more robust and user-friendly, so keep practicing and you'll get the hang of it in no time.

■ **Q2. Write a Python class to represent a bank account using object-oriented programming principles.**

****Input:**** The initial balance and a list of transactio...

Your Answer:

__SKIPPED__

Correct Answer:

```
class BankAccount:
    def __init__(self, initial_balance):
        self.balance = initial_balance

    def deposit(self, amount):
        self.balance += amount

    def withdraw(self, amount):
        if amount > self.balance:
            print("Insufficient funds")
        else:
            self.balance -= amount

    def get_balance(self):
        return self.balance

def calculate_final_balance(initial_balance, transactions):
    account = BankAccount(initial_balance)
    for transaction in transactions:
        if transaction[0] == "deposit":
            account.deposit(transaction[1])
        elif transaction[0] == "withdraw":
            account.withdraw(transaction[1])
    return account.get_balance()
```

```

initial_balance = 100
transactions = [("deposit", 100), ("withdraw", 50), ("deposit", 200)]
final_balance = calculate_final_balance(initial_balance, transactions)
print("Initial Balance:", initial_balance)
print("Transactions:", transactions)
print("Final Balance:", final_balance)

```

■ It appears that the student's code was not provided, but based on the correct code, it's likely that the issue was with the implementation of the BankAccount class and the calculate_final_balance function. The correct code works because it properly defines a BankAccount class with methods to deposit, withdraw, and get the balance, and a function to calculate the final balance after a series of transactions, allowing for a clear and organized simulation of a bank account. By using object-oriented programming principles, the correct code provides a structured and easy-to-understand solution to the problem, making it a great example to learn from.

■ **Q3. Write a Python program to convert a CSV file to a JSON file. ****Input:**** The name of the CSV file (e.g., "data.csv") ****Output:**** The name of the result...**

Your Answer:

__SKIPPED__

Correct Answer:

```

import csv
import json

def csv_to_json(csv_file_name):
    # Read the CSV file
    with open(csv_file_name, 'r') as csv_file:
        csv_reader = csv.DictReader(csv_file)
        data = list(csv_reader)

    # Get the JSON file name by replacing '.csv' with '.json'
    json_file_name = csv_file_name.replace('.csv', '.json')

    # Write the data to the JSON file
    with open(json_file_name, 'w') as json_file:
        json.dump(data, json_file, indent=4)

    print(f"CSV file '{csv_file_name}' has been converted to JSON file '{json_file_name}'")

# Test the function
csv_file_name = "data.csv"
csv_to_json(csv_file_name)

# Read and print the JSON file
with open(csv_file_name.replace('.csv', '.json'), 'r') as json_file:
    json_data = json.load(json_file)
    print("JSON Data:")
    print(json.dumps(json_data, indent=4))

```

■ It appears that the student's code was not provided, but a common issue in this type of task is that the student may not have properly read the CSV file using `csv.DictReader` or may have failed to replace the CSV file extension with a JSON file extension. The correct code works because it uses `csv.DictReader` to correctly parse the CSV file into a list of dictionaries, and then uses the `json.dump` function to write this data to a new JSON file with the correct file extension. By following this approach, you can ensure a successful conversion from CSV to JSON, so keep practicing and you'll become proficient in working with different file formats in Python!

■ **Q4. Write a Python class to manage student information. The class should have methods to add, remove, and display student records. ****Input:**** A list of st...**

Your Answer:

__SKIPPED__

Correct Answer:

```

class StudentManager:
    def __init__(self):
        self.students = []

    def add_student(self, student):
        self.students.append(student)
        print("Student added successfully.")

    def remove_student(self, name):
        for student in self.students:
            if student["name"] == name:
                self.students.remove(student)
                print("Student removed successfully.")
                return
        print("Student not found.")

    def display_students(self):
        if not self.students:
            print("No student records found.")
        else:
            for student in self.students:
                print(f"Name: {student['name']}, Age: {student['age']}")

    def update_students(self, students):
        self.students = students
        print("Student records updated.")

def main():
    manager = StudentManager()
    students = [{"name": "John", "age": 20}, {"name": "Jane", "age": 22}]
    manager.update_students(students)
    manager.display_students()
    manager.remove_student("John")
    manager.display_students()

if __name__ == "__main__":
    main()

```

■ The issue with the student's code is that it was not provided, but based on the correct code, it seems that the student may have struggled with defining a class and its methods to manage student records effectively. The correct code works because it defines a `StudentManager` class with clear methods for adding, removing, and displaying student records, allowing for efficient management of student information. By following this structure, you can create a robust and organized system for handling student data, and with practice, you'll become more proficient in designing and implementing classes in Python.

■ **Q5. Write a Python function to calculate the salary of an employee given their base salary and bonus percentage. ***Input:** The base salary and bonus perc...***

Your Answer:

dfsdg

Correct Answer:

```

def calculate_total_salary(base_salary, bonus_percentage):
    bonus_amount = (bonus_percentage / 100) * base_salary
    total_salary = base_salary + bonus_amount
    return total_salary

base_salary = 5000
bonus_percentage = 10

total_salary = calculate_total_salary(base_salary, bonus_percentage)

```

```
print("Base Salary: ", base_salary)
print("Bonus Percentage: ", bonus_percentage, "%")
print("Total Salary: ", total_salary)
```

■ It looks like there's an issue with your code - it appears to be a jumbled collection of characters (`disdg`) rather than a coherent Python function. Don't worry, we can start fresh! The correct code works by defining a function `calculate\_total\_salary` that takes `base\_salary` and `bonus\_percentage` as inputs, calculates the `bonus\_amount` by multiplying the `base\_salary` with the `bonus\_percentage` divided by 100, and then returns the `total\_salary` by adding the `base\_salary` and `bonus\_amount`. With a clear and logical structure, this function accurately calculates the total salary, and I'm happy to help you understand it better.

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.